

Clibrator Experiment Plan

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1 Science Experiment Plan

Before the formal Science Run, all the stations must make sure that the function of the calibrator is normal. The detail can be seen in early file "Calibrator_Test_Guidance.pdf".

1.1 Experiment Framework

In view of the current facility conditions, the overall experimental plan remains consistent with past Science Runs, with the primary distinction being the arrangement of signal injections. The specific plan is outlined as follows:

- Before the Science Run, we have to agree upon a **communication time**, **every station must ensure that it can receive the information from Beijing at the communication time**. This time can be chosen to be convenient for every station, like UTC 17:00:00.
- The entire Science Run is planned to last 7 —10 days (take 7 days for example in the following context). During the Science Run, **only 1 Domain Wall event is injected into the network everyday**. Hence, totally 7 Domain Wall events will be injected.
- To ensure the smooth execution of the experiment, **all the information of the total 7 injections will be prepared in advance in Beijing before the Science Run**, which includes
 1. The effective vector field $\mathbf{m}$ associated with the Domain Wall event.
 2. The relative velocity between Domain Wall and Earth $\mathbf{v}_{re}$ (We can just change the amplitude of the injected signal, the width of the signal can not be changed during the Science Run, so actually we can only change the direction of $\mathbf{v}_{re}$ during one single Science Run).
 3. Domain Wall's trigger (arrival) time at Beijing, noted by t_1 .
 4. Based on these parameters, we can calculate the signal amplitude s_i at every station, as well as the trigger times t_i for other stations.
- **At the communication time, Beijing will send today's injection information to other stations:**
 1. The injection amplitudes s_i (in the unit of pT) will not be encrypted, each station can change the amplitude using the LabView Program "Magnetic calibration control system".
 2. **The injection time t_i will be encrypted.** Each station inputs the encrypted injection time into the GPS trigger program, which then utilizes pre-embedded logic to decrypt the trigger timestamps. This protocol ensures that, without manual access to the program's output, other stations remain unaware of the daily schedule, while the system is capable of autonomously firing the DW signals at the designated times.

1.2 Things to be determined

- When to start? How long?
- A communication time.
- What kinds of event will be injected?
- Any other items.

We can brainstorm these in Wednesday's meeting.